

PROGRESS REPORT 1: Implementation Plan of Group 3

University Library & Research Management System

Course: Database Systems

Group: 3 – Library Management

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Date: 06 May 2026

1. Introduction

This project focuses on the design and implementation of a University Library & Research Management System, intended to model a real-world academic environment. Unlike traditional library systems that primarily handle book borrowing and inventory management, this system extends its functionality to include academic research publications, citation tracking, and textbook-course integration.

The objective of this first progress report is to present the implementation plan, including system scope, initial database design, and insights gathered from real-world research.

2. Research and Real-World Insights

To ensure that the system reflects real-world practices, interviews were conducted with professionals from three different domains:

- Technical University of Berlin Library
- Friedrichshain Public Library
- A local bookstore representative

These interviews helped identify operational differences and requirements across different systems.

2.1 University Libraries

University libraries focus heavily on academic and research support. Their systems include:

- Integration with courses, departments, and faculty
- Access control for restricted materials (e.g., journals, theses)
- Research publication storage
- Citation tracking and academic referencing

2.2 Public Libraries

Public libraries prioritize community access and general reading. Their systems are simpler and include:

- Basic borrowing and returning processes
- Broad categorization (fiction, media, children's books)
- No research or citation tracking
- Age Restriction for above PG 13 content

2.3 Bookstores (Private Systems)

Bookstores operate on a commercial model, focusing on:

- Sales and transactions
- Pricing and stock management
- Customer purchase tracking

Unlike libraries, bookstores do not support borrowing or reservations.

Conclusion from Research

The findings revealed the need for a **hybrid system** that combines:

- Traditional library operations
 - Academic research management
 - Educational integration
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3. System Scope and Architecture

Based on the research findings, the system is designed as a **multi-module architecture** consisting of three main components:

3.1 Library Management Module

Handles core library operations such as:

- Book and copy management
- Borrowing and returning processes
- Reservation handling
- Fine calculation for overdue items

3.2 Research Management Module

Supports academic research activities, including:

- Storage of publications (journals, theses, conference papers)
- Author relationships
- Citation tracking
- Digital access control

3.3 Academic Integration Module

Links academic workflows with library resources:

- Mapping textbooks to courses
- Associating materials with professors and semesters

This modular structure ensures scalability and reflects real-world usage.

4. Initial Database Design

The database design follows normalization principles up to the **Third Normal Form (3NF)** to ensure data integrity and reduce redundancy.

4.1 Core Entities

- **Users** – stores student, faculty, and staff information
- **Books** – bibliographic details
- **Copies** – physical instances of books
- **Borrow_Records** – borrowing transactions
- **Reservations** – reservation queues
- **Fines** – penalties for overdue items

4.2 Research Entities

- **Publications**
- **Authors**
- **Publication_Authors**
(many-to-many relationship)
- **Citations** (recursive relationship)
- **Research_Access**

4.3 Academic Entities

- **Textbook_Courses** – links textbooks to courses, professors, and semesters

5. Key Design Decisions

Several important design decisions were made:

- **Separation of Books and Copies**
Allows multiple physical copies of a single book

- **Many-to-Many Relationships**
Implemented using junction tables for flexibility
- **Citation System**
Enables tracking of references between publications
- **Academic Integration**
Links textbooks with courses and departments

6. Implementation Plan

The project will follow a phased implementation approach:

- **Phase 1 (Completed):**
Research, interviews, system design, and entity identification
- **Phase 2 (In Progress):**
ERD creation, schema design, normalization
- **Phase 3 (Upcoming):**
SQL implementation, sample data insertion, query development
- **Phase 4 (Final Stage):**
Testing, validation, and documentation

7. Interview Questions and Requirements Gathering

To better understand real-world systems, the following questions were used during interviews:

7.1 Library Operations

- What is your role in the library?
- How many users (students/staff) does the library serve?

- Approximately how many resources are managed?
- What are the most common daily tasks?

7.2 Borrowing and Circulation

- How are books issued and returned?
- What information is recorded when a book is borrowed?
- How are due dates tracked?
- Do users reserve books in advance? If yes, how?

7.3 Inventory and Book Management

- How are new books added to the system?
- What details are stored for each book (ISBN, author, category)?
- Are multiple copies of the same book tracked?
- How are lost or damaged books handled?

7.4 User Management

- What types of users exist (students, teachers, external members)?
- What information is stored about each user?
- Are there borrowing limits?
- Do different users have different privileges?
- What reports are generated regularly?

7.5 System Usage and Challenges

- Do you monitor overdue books?
- How do users search for books?
- What filters are important (author, title, category, year)?

- Do users check availability before visiting?
- What challenges exist in the current system?
- Are there manual processes that take too long?
- Do you lose track of books or records?
- What errors occur frequently?

8. Additional Recommended Questions

To further improve the system design, the following additional questions are recommended:

- What are the most time-consuming tasks in the current system?
- Is the current system fully digital or partially manual?
- What software or tools are currently used?
- What common complaints do users have?
- What additional features do users request?
- How is access to restricted materials controlled?
- Is there any authentication or role-based access system?
- How are research publications currently managed?
- Is citation tracking implemented in the current system?

9. Challenges Faced

During this phase, several challenges were encountered:

- Balancing system simplicity with real-world complexity
 - Designing efficient many-to-many relationships
 - Modeling citation relationships without redundancy
 - Integrating library, research, and academic modules into one system
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10. Next Steps

The next phase will focus on:

- Finalizing the ERD diagram
 - Writing complete SQL schema definitions
 - Implementing sample queries
 - Enhancing system functionality
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11. Conclusion

This phase established a strong foundation for a scalable and realistic **University Library & Research Management System**. By incorporating real-world insights and extending beyond traditional designs, the system effectively reflects the complexity of modern academic environments.

The next stages will focus on implementing the design into a fully functional database system.
